








Advanced Time Series Analysis and Forecasting

Chapter 11: Spectral and wavelet analysis



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Acronyms used in this material (1/2)

- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **ARMA**: AutoRegressive Moving Average (model)
- **BC**: Breitung–Candelon (frequency-domain causality test)
- **BET**: Bucharest Exchange Trading (index)
- **BK**: Baxter–King (band-pass filter)
- **BUX**: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- **CF**: Christiano–Fitzgerald (band-pass filter)
- **CFR**: Croux, Forni and Reichlin (2001)
- **COI**: Cone Of Influence (of a wavelet transform)
- **CWT**: Continuous Wavelet Transform
- **DCC-GARCH**: Dynamic Conditional Correlation GARCH
- **DFT**: Discrete Fourier Transform
- **DPSS**: Discrete Prolate Spheroidal Sequences (Slepian tapers)
- **DWT**: Discrete Wavelet Transform
- **EODHD**: EOD Historical Data (market-data provider)
- **FFT**: Fast Fourier Transform
- **FRED**: Federal Reserve Economic Data

Acronyms used in this material (2/2)

- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GDP**: Gross Domestic Product
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HICP**: Harmonised Index of Consumer Prices
- **HP**: Hodrick–Prescott (filter)
- **LA**: Least Asymmetric (Daubechies wavelet filter)
- **LLM**: Large Language Model
- **MA**: Moving Average
- **MODWT**: Maximal Overlap Discrete Wavelet Transform
- **MRA**: MultiResolution Analysis
- **MSE**: Mean Squared Error
- **PX**: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- **QS**: Quadratic Spectral (kernel)
- **RMSE**: Root Mean Squared Error
- **SSA**: Singular Spectrum Analysis
- **SVD**: Singular Value Decomposition
- **TSA**: Time Series Analysis
- **US**: United States
- **VAR**: Vector AutoRegression
- **VARMA**: Vector AutoRegressive Moving Average (model)
- **WIG20**: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Today's question and route

- **Question**: how is the variance of a time series, and the co-movement of two series, distributed over frequencies, and how does this distribution change over time?
  - cycles of different lengths answer different economic questions: the business cycle, seasonality, the reaction of markets within a week
- **Route** of the chapter
  - the spectral representation; estimation theory: consistency, bandwidth, multitaper
  - two series: coherence, phase, dynamic correlation, frequency-domain causality
  - band-pass filters and Hamilton's critique of the HP filter; Romania and the euro area
  - evolutionary spectra; wavelets: MODWT, variance and correlation by scale, wavelet coherence
- We build on TSA, Chapter 12 (periodogram, smoothing, Fisher's test, coherence), Chapter 6 (state space) and Chapter 10 (the spectral pole of long memory); Seminar 11 comes before this lecture

Learning outcomes

- State the spectral representation of a stationary process and derive the spectrum of a filtered process
- Derive the bias and variance of lag-window estimators, choose the bandwidth, and build multitaper estimates with valid confidence bands
- Estimate and test coherence, phase, gain and dynamic correlation, and test Granger causality at a given frequency
- Evaluate trend–cycle filters by their gain, and measure business-cycle synchronisation
- Decompose variance and correlation by scale with the MODWT and test wavelet coherence by Monte Carlo

Reading, data and tools

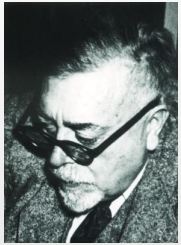
- Spectral theory and estimation: Brockwell and Davis (1991); Percival and Walden (1993); Shumway and Stoffer (2017); Thomson (1982)
  - Wavelets: Percival and Walden (2000); Torrence and Compo (1998); Grinsted, Moore and Jevrejeva (2004); Aguiar-Conraria and Soares (2014)
  - Business cycles: Baxter and King (1999); Christiano and Fitzgerald (2003); Hamilton (2018); Croux, Forni and Reichlin (2001)
  - Survey of forecasting practice: Petropoulos et al. (2022)
- Python Quantlets of this chapter: Quantlets/Ch_11
  - spectra, multitaper, cross-spectra, Geweke and Breitung–Candelon, filters, MODWT, Morlet transform and wavelet coherence written in `numpy/scipy`; `PyWavelets` gives the wavelet filters
- Lecture notebook: open in Google Colab

Data used in this chapter

Series	Source	Sample
Real GDP: Romania, euro area, Poland, Hungary, Czechia	Eurostat, chain-linked volumes, seasonally and calendar adjusted, quarterly	1995Q1 – 2026Q2
Industrial production: Romania, euro area	Eurostat, monthly, adjusted and unadjusted	January 2000 – July 2026
HICP annual inflation: Romania, Poland, euro area	Eurostat, monthly	January 2001 – August 2026
BET, DAX, S&P 500	EODHD daily closes; weekly returns from Friday closes	2000 – 2026
Brent crude oil	FRED (US Energy Information Administration), daily	2000 – 2026

- Growth rates and returns are 100 times log differences; cycles are in % of trend; markets are aligned on common trading days

From harmonic analysis to time series



Norbert Wiener

Photo: Konrad Jacobs, MFO; CC BY-SA 2.0 de; Wikimedia Commons

- 1930: Wiener's generalised harmonic analysis; 1934: Khinchin; autocovariance and spectrum are a Fourier pair
- 1942: Cramér's representation of a stationary process as a random superposition of sinusoids
- 1961–1966: lag windows Parzen (1961); the typical spectral shape of economic variables Granger (1966)
- 1982: multitaper estimation Thomson (1982); frequency-domain causality Geweke (1982)
- 1984–1989: wavelets Grossmann and Morlet (1984), Daubechies (1988), Mallat (1989); 1998: the practical guide Torrence and Compo (1998)

Known from TSA and new here

- Known (TSA, Chapter 12): Fourier frequencies, the spectrum of ARMA models, the periodogram and its χ^2_2 law, Daniell smoothing, tapering, Fisher's test, a first look at coherence and wavelets
- New: the research toolkit
  - asymptotic theory of spectral estimators: bias, variance, optimal bandwidth, valid confidence bands, multitaper
  - inference on co-movement: coherence thresholds, phase errors, dynamic correlation, causality by frequency
  - filters judged by their gain; time–frequency analysis with significance tests that control false discoveries
- Case studies, with the specifications of the papers, on our data
  - spectral: Thomson (1982); Croux, Forni and Reichlin (2001); Breitung and Candelon (2006)
  - filters and wavelets: Baxter and King (1999); Cogley and Nason (1995); Hamilton (2018); Torrence and Compo (1998); Grinsted, Moore and Jevrejeva (2004)

Autocovariance and spectral distribution (1/2)

- **Herglotz** Brockwell and Davis (1991): $\gamma(\cdot)$ is the autocovariance of a stationary process if and only if

$$\gamma(h) = \int_{(-\pi,\pi]} e^{i\omega h}\, dF(\omega)$$

  - $\gamma(h) = \mathrm{Cov}(X_{t+h}, X_t)$; ω: angular frequency in radians per period; F: the **spectral distribution**, non-decreasing, right-continuous and bounded
- If $\sum_h |\gamma(h)| < \infty$, F has a density, the **spectral density**

$$dF(\omega) = f(\omega)\, d\omega, \qquad f(\omega) = \frac{1}{2\pi} \sum_h \gamma(h) e^{-i\omega h}$$

Autocovariance and spectral distribution (2/2)

- At $h = 0$: the spectrum is a **decomposition of variance by frequency**

$$\gamma(0) = \mathrm{Var}(X_t) = \int_{-\pi}^{\pi} f(\omega)\, d\omega$$

  - $f(\omega)\, d\omega$: the variance contributed by cycles with frequency near ω; period $2\pi/\omega$
  - quarterly data: the business-cycle band of 6–32 quarters is $\omega \in [\pi/16, \pi/3]$
- A jump of F at $\pm\omega_0$ is a deterministic cycle (a **line spectrum**); long memory is a pole of f at $\omega = 0$ (Chapter 10)

Cramér's representation



Harald Cramér, 1951

Photo: unknown photographer, Svenska Dagbladet (1951); public domain; Wikimedia Commons

- Every zero-mean stationary process is a sum of sinusoids with random amplitudes Brockwell and Davis (1991)

$$X_t = \int_{(-\pi,\pi]} e^{i\omega t}\, dZ(\omega)$$

  - $dZ(\omega)$: the random amplitude at frequency ω; Z has **orthogonal increments**: $\mathbb{E}[dZ(\omega)\overline{dZ(\lambda)}] = 0$ for $\omega \neq \lambda$, $\mathbb{E}|dZ(\omega)|^2 = dF(\omega)$
- Random amplitudes at different frequencies are uncorrelated: the frequency components can be studied one by one
- The discrete Fourier transform (DFT) at the Fourier frequencies $\omega_j = 2\pi j/n$

$$d(\omega_j) = n^{-1/2} \sum_{t=1}^{n} x_t e^{-i\omega_j t}$$

  - the sample analogue of $dZ(\omega_j)$: nearly uncorrelated across Fourier frequencies

Linear filters in the frequency domain

- A linear filter multiplies each frequency component by the **transfer function** $A(\omega)$

$$Y_t = \sum_j a_j X_{t-j} \;\Rightarrow\; dZ_Y(\omega) = A(\omega)\, dZ_X(\omega), \quad A(\omega) = \sum_j a_j e^{-i\omega j}$$

$$f_Y(\omega) = |A(\omega)|^2 f_X(\omega)$$

  - a_j: filter weights, $\sum_j |a_j| < \infty$; $|A(\omega)|$: the **gain**; $\arg A(\omega)$: the **phase shift**
  - symmetric weights $a_j = a_{-j}$: $A(\omega)$ is real, no phase shift; the property every trend–cycle filter wants
- First difference: $|1 - e^{-i\omega}|^2 = 2(1 - \cos\omega)$, zero at $\omega = 0$
  - differencing removes the long cycles and amplifies the short ones
- Slutsky effect: a moving sum of white noise has a peak in its spectrum, i.e. **cycles created by the filter**; the HP critique of Section 4 is the same phenomenon

The spectral density matrix (1/2)

- Vector process X_t of dimension k: the Fourier transform of the autocovariance matrices

$$f(\omega) = \frac{1}{2\pi} \sum_h \Gamma(h) e^{-i\omega h}, \qquad \Gamma(h) = \mathrm{Cov}(X_{t+h}, X_t)$$

  - $f(\omega)$: a $k \times k$ Hermitian ($f = f^*$, the conjugate transpose) non-negative definite matrix at each frequency
- VARMA $\Phi(L)X_t = \Theta(L)\varepsilon_t$: the multivariate analogue of the ARMA spectrum

$$f(\omega) = \frac{1}{2\pi} \Phi(e^{-i\omega})^{-1} \Theta(e^{-i\omega}) \, \mathbf{\Sigma} \, \Theta(e^{-i\omega})^* \Phi(e^{-i\omega})^{-*}$$

  - Φ, Θ: matrix AR and MA polynomials; $\mathbf{\Sigma}$: covariance of the innovations ε_t; $^{-*}$: inverse of the conjugate transpose

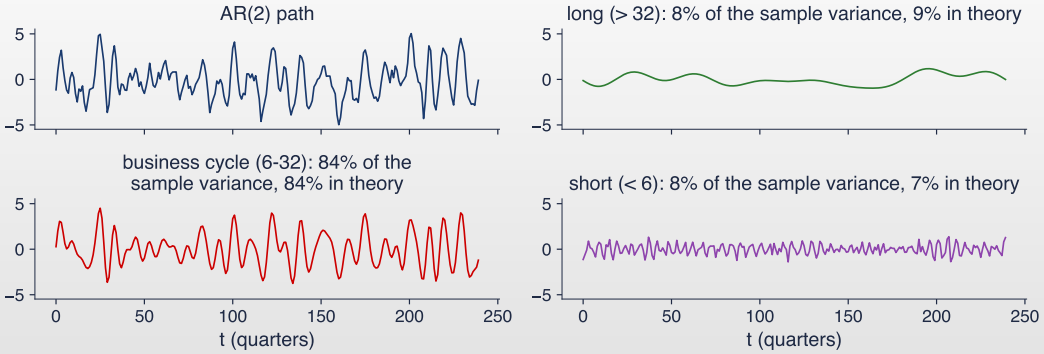
The spectral density matrix (2/2): cross-spectra

- Off-diagonal elements: the **cross-spectrum**, complex-valued

$$f_{xy}(\omega) = c_{xy}(\omega) - i\, q_{xy}(\omega)$$

  - c_{xy}: co-spectrum, even in ω (contemporaneous co-movement), $\int c_{xy} = \mathrm{Cov}(X_t, Y_t)$
  - q_{xy}: quadrature spectrum, odd in ω (lead–lag between the series)
- Section 2 estimates the diagonal, Section 3 the off-diagonal, Section 4 factorises f to measure causality

Variance by frequency band



- An AR(2) path, $X_t = 1.3X_{t-1} - 0.7X_{t-2} + \varepsilon_t$, 240 "quarters", split by the DFT into three bands; shares of variance in the sample and $2\int_{\text{band}} f/\gamma(0)$ in theory

ATS_ch11_spectral_estimation

Interpreting the band decomposition

- The spectral peak is at a period of 9.5: 84% of the variance lies in the band of 6–32 periods (sample: 84%)
- Long and short cycles carry 9% and 7% in theory: the three components are uncorrelated by Cramér's theorem, so the shares add up
- The "business cycle" of an AR(2) is a stochastic pseudo-cycle: irregular amplitude and length, no line in the spectrum
- Band-pass filters (Section 4) estimate exactly this middle component from a finite sample

Recap: The spectral representation

- The spectrum decomposes variance by frequency; Cramér: uncorrelated random amplitudes at each frequency
- A linear filter multiplies the spectrum by its squared gain; symmetric filters do not shift phase
- The spectral matrix contains cross-spectra: co-movement and lead–lag by frequency

Limitations of the periodogram

- For a linear process the periodogram ordinates are asymptotically scaled χ^2_2 variables Brockwell and Davis (1991)

$$X_t = \sum_j \psi_j \varepsilon_{t-j} \;\Rightarrow\; I(\omega_j) \Rightarrow f(\omega_j)\,\frac{\chi^2_2}{2}$$

  - ψ_j: weights with $\sum_j |j|^{1/2}|\psi_j| < \infty$; ε_t: i.i.d. innovations; $f > 0$; asymptotically independent over fixed Fourier frequencies ω_j
  - $\mathbb{E}I(\omega_j) \to f(\omega_j)$ but $\mathrm{Var}I(\omega_j) \to f(\omega_j)^2$: **inconsistent**, the variance does not fall with n
  - derivation for white noise and for a linear process: Appendix
- Leakage: the mean periodogram is the true spectrum smoothed by the Fejér kernel F_n

$$\mathbb{E}I(\omega) = \int F_n(\omega - \lambda) f(\lambda)\, d\lambda$$

  - the side lobes of F_n decay only like $1/(n\omega^2)$, so power **leaks** from strong to weak frequencies
- Two remedies: **average** neighbouring frequencies (consistency) and **taper** the data (bias); multitaper does both

Lag-window estimators (1/2)

- Down-weight the noisy sample autocovariances at long lags Parzen (1961)

$$\hat{f}(\omega) = \frac{1}{2\pi} \sum_{|h|<n} k(h/M)\, \hat{\gamma}(h) e^{-i\omega h}$$

  - $\hat{\gamma}(h)$: sample autocovariance; k: the **lag window**, $k(0) = 1$, decreasing in $|u|$; M: the truncation (bandwidth) parameter

- Equivalently: a weighted average of the periodogram with the **spectral window** W_M

$$\hat{f}(\omega) = \int W_M(\omega - \lambda) I(\lambda)\, d\lambda, \qquad W_M(\omega) = \frac{1}{2\pi} \sum_h k(h/M) e^{-i\omega h}$$

  - a large M gives a narrow W_M: little smoothing

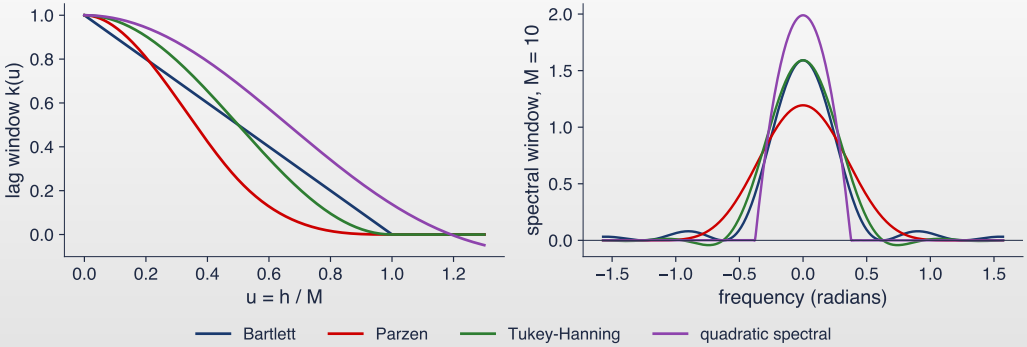
Lag-window estimators (2/2): the kernels

- Bartlett $k(u) = 1 - |u|$; Parzen; Tukey–Hanning; quadratic spectral (QS): the same kernels as the HAC estimators of Chapter 0 at $\omega = 0$
  - a non-negative W_M guarantees $\hat{f} \geq 0$ (Bartlett, Parzen, QS); Tukey–Hanning can go negative
- Characteristic exponent q: how flat the kernel is at zero

$$k_q = \lim_{u \to 0} \frac{1 - k(u)}{|u|^q}$$

  - Bartlett $q = 1$, $k_1 = 1$; Parzen $q = 2$, $k_2 = 6$; QS $q = 2$, $k_2 = 1.42$; a larger q means smaller bias (next slides)

Lag windows and spectral windows



⊡ Left: $k(u)$; right: the spectral window $W_M(\omega)$ for $M = 10$, the weights that each estimator gives to the periodogram around ω

ATS_ch11_spectral_estimation

Interpreting the windows

- A narrow W_M means low bias and high variance
  - at $M = 10$, Parzen is the widest (its $\int k^2 = 0.54$ is the smallest) and QS the narrowest
- Bartlett has side lobes (leakage through the window itself); Tukey–Hanning has negative weights
- Comparing kernels at the same M is misleading: compare them at the same variance (the same equivalent degrees of freedom)
- The QS kernel is optimal in MSE among kernels with non-negative estimates Andrews (1991)

Bias, variance and the optimal bandwidth (1/2)

- If $M \to \infty$ and $M/n \to 0$: $\hat{f}(\omega) \to_p f(\omega)$ (**consistency**) Parzen (1961); variance and bias

$$\mathrm{Var}\hat{f}(\omega) \approx \frac{M}{n} f(\omega)^2 \int k^2(u)\, du, \qquad \mathbb{E}\hat{f}(\omega) - f(\omega) \approx -k_q M^{-q} f^{(q)}(\omega)$$

  - the variance grows with M (doubled at $\omega = 0, \pm\pi$); the bias falls with M
  - $f^{(q)}(\omega) = \frac{1}{2\pi} \sum_h |h|^q \gamma(h) e^{-i\omega h}$: a generalised q-th derivative of f, large where the spectrum is curved (peaks)

Bias, variance and the optimal bandwidth (2/2)

- Minimising MSE = squared bias + variance over M

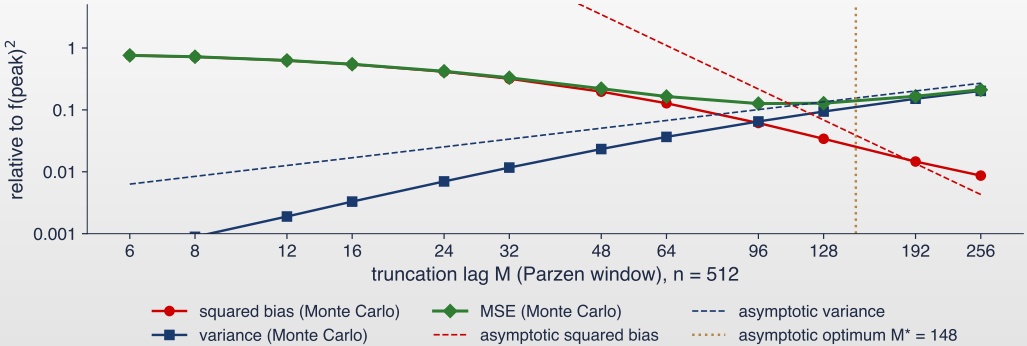
$$M^* = \Big(\frac{q k_q^2 f^{(q)}(\omega)^2\, n}{f(\omega)^2 \int k^2}\Big)^{1/(2q+1)} \propto n^{1/(2q+1)}, \qquad \text{MSE} \propto n^{-2q/(2q+1)}$$

  - $q = 2$ kernels converge faster ($n^{-4/5}$) than Bartlett ($n^{-2/3}$)
- Inference: a scaled χ^2 with equivalent degrees of freedom ν

$$\frac{\nu \hat{f}(\omega)}{f(\omega)} \approx \chi^2_\nu, \qquad \nu = \frac{2n}{M \int k^2}$$

  - the confidence band $[\nu\hat{f}/\chi^2_{\nu,0.975}, \nu\hat{f}/\chi^2_{\nu,0.025}]$ has constant width in log scale

The bias–variance trade-off at a spectral peak



- Parzen estimator at the peak (period 11.1) of $X_t = 1.6X_{t-1} - 0.9X_{t-2} + \varepsilon_t$, $n = 512$, 400 simulations; curves divided by $f(\text{peak})^2$; dashed: the asymptotic formulas

ATS_ch11_spectral_estimation

Interpreting the trade-off

- With $M = 12$ the relative RMSE is 0.79, almost all of it bias (0.79): heavy smoothing flattens the peak
- The Monte Carlo optimum is $M = 96$ (relative RMSE 0.36); the asymptotic formula gives $M^* = 148$: same order, but the formula ignores higher-order bias terms
- The asymptotic variance is accurate only once the window is narrower than the peak; the asymptotic bias explodes for small M
- A sharp spectral peak needs a much larger M than the HAC rule for $\omega = 0$ would suggest: bandwidth is local

Choosing the bandwidth

- **Plug-in** Andrews (1991): approximate $f^{(q)}/f$ by a parametric (AR(1)) model, then use M^*

$$M = 1.1447\,(\hat{\alpha}(1)\,n)^{1/3}\ \text{(Bartlett)}, \qquad M = 1.3221\,(\hat{\alpha}(2)\,n)^{1/5}\ \text{(QS)}, \qquad \omega = 0$$

  - $\hat{\alpha}(q)$: the ratio $(f^{(q)}(0)/f(0))^2$ computed from the fitted AR(1), e.g. $\hat{\alpha}(2) = 4\hat{\rho}^2/(1-\hat{\rho})^4$ ($\hat{\rho}$: AR coefficient)
  - good for long-run variances (Chapter 0); poor near sharp peaks, where the AR(1) approximation is wrong
- **Cross-validation** Hurvich (1985): minimise over M a leave-one-out Whittle criterion

$$\mathrm{CV}(M) = \sum_j \Big[\log \hat{f}_{-j}(\omega_j) + \frac{I(\omega_j)}{\hat{f}_{-j}(\omega_j)} \Big]$$

  - $\hat{f}_{-j}$: the smoothed estimate computed without the ordinate $I(\omega_j)$
- **Resolution**: two peaks closer than the bandwidth (about $1/M$ cycles) merge; decide the resolution you need before looking at the data
  - report the estimate for two or three bandwidths: conclusions that depend on M are not conclusions

Multitaper estimation (1/2): Slepian tapers

- **Slepian (DPSS) tapers** $v^{(k)}$, $k = 0, \dots, K-1$ Thomson (1982)
  - orthonormal weight sequences of length n that maximise the share λ_k of their spectral energy inside $[-W, W]$; W: half-bandwidth in cycles
  - about $2nW$ of them have $\lambda_k \approx 1$; usually $K = 2NW - 1$ with $NW = nW$ (e.g. $NW = 4$, $K = 7$)
- Eigenspectra: one tapered periodogram per taper; the estimate is their average

$$\hat{S}_k(\omega) = \frac{1}{2\pi}\Big|\sum_t v_t^{(k)} x_t e^{-i\omega t}\Big|^2, \qquad \hat{f}^{\mathrm{mt}}(\omega) = \frac{1}{K}\sum_{k=0}^{K-1} \hat{S}_k(\omega)$$

  - the $\hat{S}_k$ are nearly uncorrelated: $2K\hat{f}^{\mathrm{mt}}/f \approx \chi^2_{2K}$; variance f^2/K, bias controlled by the taper concentration

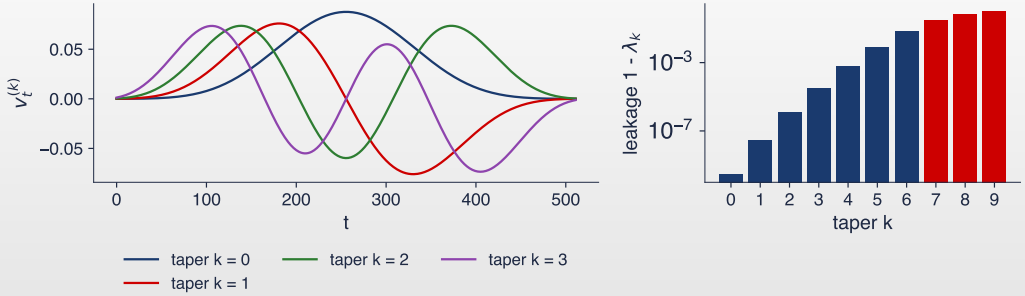
Multitaper estimation (2/2): adaptive weights

- **Adaptive weights** Percival and Walden (1993): down-weight leaky tapers where f is small

$$d_k(\omega) = \frac{\sqrt{\lambda_k}\, f(\omega)}{\lambda_k f(\omega) + (1-\lambda_k)\sigma^2}$$

  - σ^2: the variance of the series; $1-\lambda_k$: the leakage of taper k; the weights depend on the unknown f, so they are iterated from $\hat{f}^{\mathrm{mt}}$
- Alternatives with the same logic: sine tapers Riedel and Sidorenko (1995); Welch's averaging of tapered segments Welch (1967)

Slepian tapers



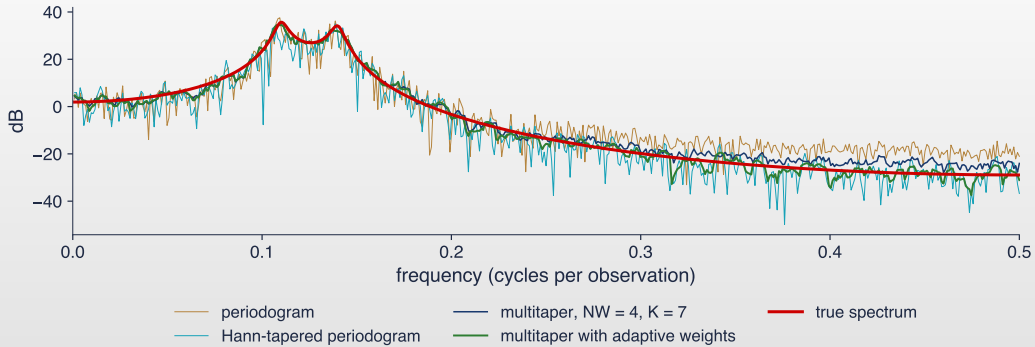
⊡ $n = 512$, $NW = 4$: the first four tapers and the leakage $1 - \lambda_k$ of the first ten (blue: $k < 2NW - 1$)

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Interpreting the tapers

- Taper k has k zero crossings: higher tapers weight the ends of the sample, so together they use all the data
- Leakage grows fast with k: $\lambda_6 = 0.937$, but $\lambda_7 = 0.70$ and $\lambda_8 = 0.30$; beyond $2NW - 1$ the tapers are useless
- The single Hann taper throws away the ends; multitapering recovers that information as extra degrees of freedom
- NW is the only tuning choice: it fixes the resolution $2W = 2NW/n$ and the variance f^2/K together

Leakage on a spectrum with a high dynamic range



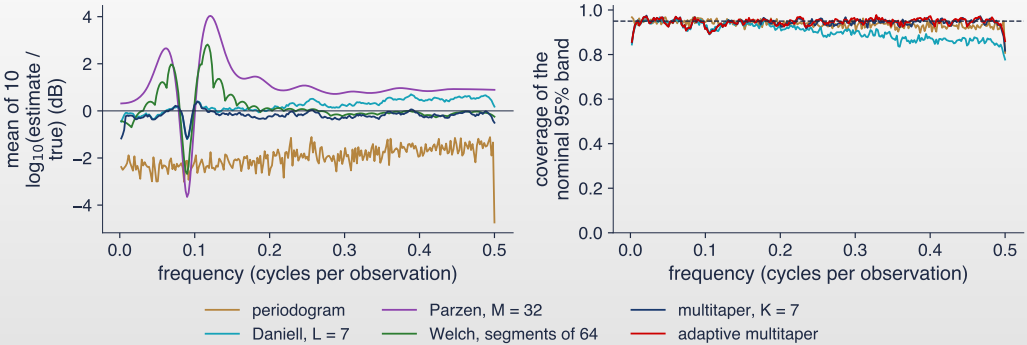
- The AR(4) of Percival and Walden (1993), $n = 1024$, range of 65 dB between peak and trough; one simulated path (median leakage of 21)

ATS_ch11_spectral_estimation

Interpreting leakage

- Mean bias at frequencies above 0.3 cycles (200 simulations): periodogram +11.2 dB, Hann taper −2.5 dB, multitaper +6.3 dB, adaptive multitaper −0.8 dB
- The raw periodogram fills the trough with power leaked from the peaks: averaging it more would not help, the bias is in each ordinate
- With equal weights the leaky high-order tapers still contaminate the trough; the adaptive weights remove it
- Economic data rarely have 60 dB ranges, but spectra of levels with a unit-root-like peak at zero do: taper or prewhiten before estimating

Five estimators in a Monte Carlo



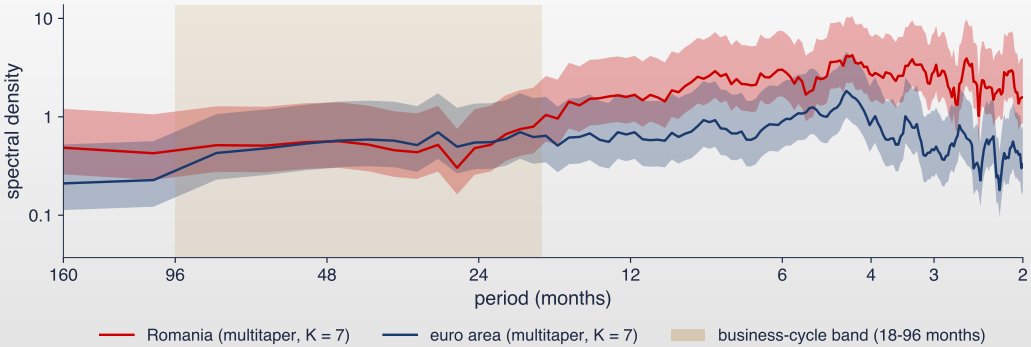
- $X_t = 1.6X_{t-1} - 0.9X_{t-2} + \varepsilon_t$, $n = 512$, 400 simulations: mean log-bias by frequency and coverage of the nominal 95% χ^2 bands

ATS_ch11_spectral_estimation

Interpreting the Monte Carlo

- Standard deviation of $10\log_{10}(\hat{f}/f)$: periodogram 5.6 dB, Daniell 1.8, multitaper 1.7, Parzen 0.9, Welch 1.1
- At the peak: Parzen -2.3 dB, Welch -1.4 dB, multitaper -0.5 dB; the periodogram's -2.0 dB is the mean of $\log \chi^2_2/2$, not a bias in levels
- Coverage of the 95% band: multitaper 94.3%, adaptive 94.6%, Daniell 90.3% (its band ignores the bias on the slopes)
- Multitaper is the only estimator here with small bias, small variance and honest bands at once

Industrial production: Romania and the euro area



- Monthly growth of industrial production (seasonally and calendar adjusted), February 2000 – July 2026, 318 months; multitaper $NW = 4$ with 95% bands

ATS_ch11_ip_spectrum

Interpreting the industrial production spectra

- Monthly growth is dominated by short periods: cycles under 6 months carry 77% of the variance in Romania and 69% in the euro area
- The business-cycle band (18–96 months) holds 2.1% (Romania) and 7.3% (euro area): growth rates hide cycles that levels show (the first-difference gain)
- Romania is more volatile (standard deviation 3.61 against 1.98), and the excess is at high frequencies, where the bands do not overlap: noise, revisions and residual calendar effects
- At periods above 12 months the two spectra cannot be told apart within the bands

Lines in the spectrum: Thomson's F test (1/2)

- Model: a sinusoid plus noise with a smooth spectrum Thomson (1982)

$$x_t = \mu \cos(\omega_0 t + \phi) + e_t$$

  - μ: amplitude; ω_0: frequency of the line; ϕ: phase; e_t: stationary noise
  - under a line, each tapered DFT is $Y_k(\omega_0) \approx \mu\, U_k(0)$, with $U_k(0) = \sum_t v_t^{(k)}$ the sum of taper k
- A regression of the eigencoefficients Y_k on the taper sums $U_k(0)$, frequency by frequency: it separates a deterministic line from a stochastic peak, which Fisher's test (TSA) cannot

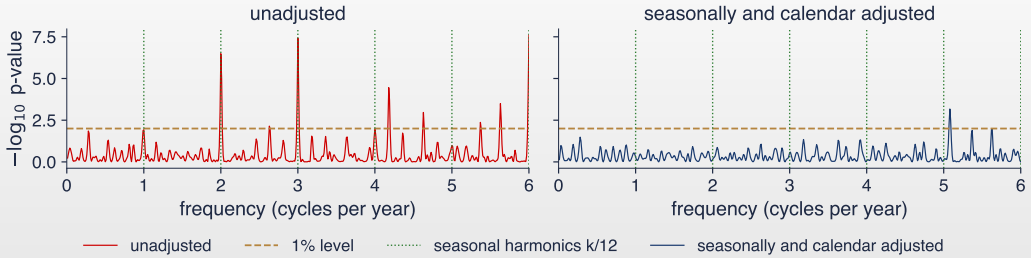
Lines in the spectrum: Thomson's F test (2/2)

- Estimated amplitude and the F statistic: explained against residual variation

$$\hat{\mu}(\omega) = \frac{\sum_k U_k(0) Y_k(\omega)}{\sum_k U_k(0)^2}, \qquad F(\omega) = \frac{(K-1)\,|\hat{\mu}|^2 \sum_k U_k(0)^2}{\sum_k |Y_k - \hat{\mu} U_k(0)|^2} \sim F_{2,2K-2}$$

  - a large $F(\omega)$: a line at ω; K: number of tapers
- Application: residual seasonality is a set of lines at $k/12$ cycles per month; trading-day effects are lines at 0.348 and 0.432 cycles per month
- Testing at hundreds of frequencies: expect about 1% false rejections at the 1% level; test only pre-specified frequencies

Residual seasonality in Romanian industrial production



- Harmonic F test ($NW = 4$, $K = 7$) on monthly growth, unadjusted and adjusted series; dotted lines: the six seasonal frequencies

ATS_ch11_ip_spectrum

Interpreting the F test

- Unadjusted: 3 of 6 seasonal harmonics are lines at 1% ($p < 0.0001$ and < 0.0001 at 2 and 3 cycles per year); the annual harmonic is borderline ($p = 0.012$)
- A strong line at 4.18 cycles per year ($p < 0.0001$): the trading-day frequency, a calendar effect, not seasonality
- Adjusted series: none of the six harmonics is significant, and there is no line at the trading-day frequency ($p = 0.22$): the adjustment passes this check
- The annual harmonic is weak because Romanian seasonality is not a pure sinusoid: its power sits at the higher harmonics

Recap: Spectral estimation theory

- The periodogram is unbiased in the limit but inconsistent and leaky; smoothing buys consistency, tapering buys low bias
- Bandwidth is a bias–variance choice with $M^* \propto n^{1/(2q+1)}$; peaks need narrow windows
- Multitaper: χ^2_{2K} bands that hold, adaptive weights against leakage, and an F test for lines

Coherence, phase and gain (1/2): coherence

- **Squared coherence**: the R^2 of the regression of $dZ_y(\omega)$ on $dZ_x(\omega)$

$$\kappa^2_{xy}(\omega) = \frac{|f_{xy}(\omega)|^2}{f_x(\omega) f_y(\omega)} \in [0, 1]$$

  - f_x, f_y: the two spectra; f_{xy}: the cross-spectrum; $\kappa^2 = 1$: y is an exact linear filter of x at frequency ω; $\kappa^2 = 0$: no linear relation at that frequency
  - invariant to filtering each series by an invertible filter: coherence does not depend on how the series are transformed (Appendix)
- **Gain** $|f_{xy}(\omega)|/f_x(\omega)$: the regression coefficient of y on x at frequency ω

Coherence, phase and gain (2/2): phase

- **Phase**: the angle of the cross-spectrum, with $\gamma_{xy}(h) = \mathrm{Cov}(x_{t+h}, y_t)$

$$\phi_{xy}(\omega) = \arg f_{xy}(\omega)$$

  - example: if $y_t = x_{t-d}$, then $f_{xy} = e^{i\omega d} f_x$, phase $\omega d > 0$: x leads by ϕ/ω periods
  - the phase is defined modulo 2π: a lead of d and a lag of $2\pi/\omega - d$ are the same phase
- Interpret the phase only where coherence is significant

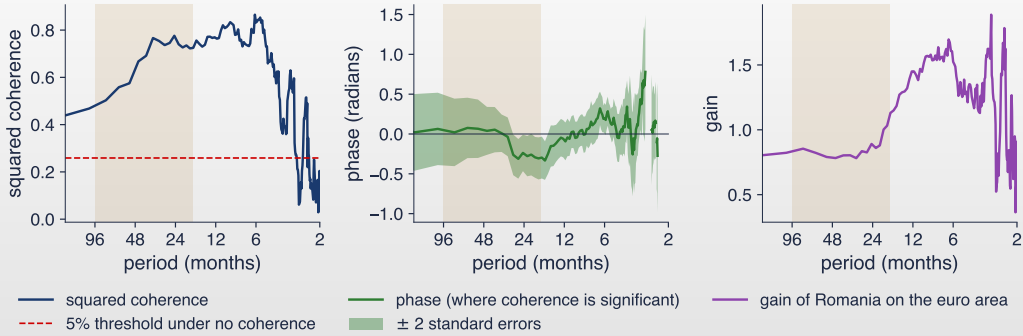
Inference on coherence and phase

- The raw cross-periodogram gives $\hat{\kappa}^2 \equiv 1$ at every frequency: coherence **must** be smoothed (here: averaged over K tapers)

$$\Pr(\hat{\kappa}^2 > c \mid \kappa^2 = 0) = (1-c)^{L-1} \;\Rightarrow\; c_{0.05} = 1 - 0.05^{1/(L-1)}$$

  - L: number of independent ordinates averaged ($L = K$ for multitaper); c: threshold; $\hat{\kappa}^2$ above $c_{0.05}$ is significant at 5%
- $\hat{\kappa}^2$ is biased upwards when κ^2 is small; confidence intervals via the Fisher transform $\tanh^{-1}\hat{\kappa}$ Shumway and Stoffer (2017)
  - phase: $\mathrm{se}(\hat{\phi}) \approx \sqrt{(1/\hat{\kappa}^2 - 1)/(2L)}$; it explodes as coherence falls
- Misalignment bias: a long delay d rotates the phase within the smoothing band and lowers $\hat{\kappa}^2$; align the series (or prewhiten) first

Romania and the euro area: coherence, phase and gain



⊡ Monthly industrial production growth, euro area (x) and Romania (y); multitaper $NW = 6$, $K = 11$; shaded: 18–96 months

ATS_ch11_cross_spectrum

Interpreting the cross-spectrum

- Coherence averages 0.69 in the business-cycle band and 0.47 below 12 months; the 5% threshold is 0.26: significant at 100% of business-cycle frequencies and 71% of short ones
- The phase is near zero: the median implied lead is 0.5 months, inside two standard errors: Romanian industry moves with the euro area within the month
- Gain 0.85 at business-cycle frequencies, 1.24 at short ones: Romania amplifies euro-area shocks only at high frequencies
- The simple correlation (0.66) mixes these bands; the coherence separates where co-movement is strong from where it is noise

Dynamic correlation

- Croux, Forni and Reichlin (2001): the correlation of the in-phase (real) components at frequency ω

$$\rho_{xy}(\omega) = \frac{c_{xy}(\omega)}{\sqrt{f_x(\omega) f_y(\omega)}} \in [-1, 1]$$

$$\kappa^2 = \rho^2 + \left(\frac{q_{xy}}{\sqrt{f_x f_y}} \right)^2$$

  - c_{xy}, q_{xy}: co-spectrum and quadrature spectrum; unlike coherence, ρ_{xy} has a **sign** and is not increased by out-of-phase movement

- Over a band Λ of frequencies

$$\rho_{xy}(\Lambda) = \frac{\int_\Lambda c_{xy}}{\sqrt{\int_\Lambda f_x \int_\Lambda f_y}}$$

  - over $[0, \pi]$ it is the ordinary correlation; over a band it equals the correlation of the band-pass filtered series (Section 4)

- **Cohesion**: a weighted average of pairwise $\rho_{xy}(\omega)$ inside a group of countries; CFR use it for euro-area business cycles

Granger causality by frequency (1/2): the Geweke measure

- Geweke (1982): in a bivariate VAR, normalise the innovations so that x's innovation is uncorrelated with the transformed y innovation; then

$$f_x(\omega) = \frac{1}{2\pi}\left(|\tilde{H}_{xx}(\omega)|^2\tilde{\sigma}_{xx} + |\tilde{H}_{xy}(\omega)|^2\tilde{\sigma}_{yy}\right)$$

  - $\tilde{H}$: the transfer function (moving-average representation) of the normalised VAR; $\tilde{\sigma}_{xx}$, $\tilde{\sigma}_{yy}$: variances of the two orthogonal innovations

- The share of $f_x(\omega)$ that comes from y

$$M_{y\to x}(\omega) = \ln\frac{2\pi f_x(\omega)}{|\tilde{H}_{xx}(\omega)|^2\tilde{\sigma}_{xx}} \geq 0$$

$$\frac{1}{\pi}\int_0^\pi M_{y\to x}(\omega)\,d\omega = \ln\frac{\sigma^2_{x|x}}{\sigma^2_{x|x,y}}$$

  - the integral is the time-domain Granger measure (under a mild condition): $\sigma^2_{x|x}$, $\sigma^2_{x|x,y}$ are the one-step forecast variances of x without and with the past of y

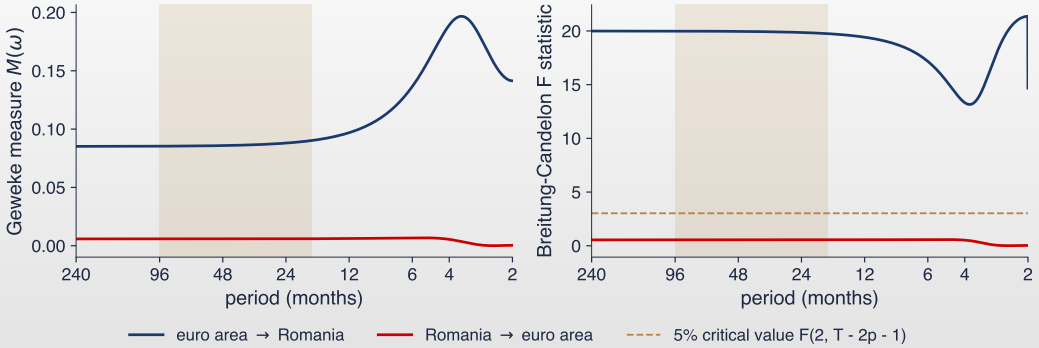
Granger causality by frequency (2/2): the Breitung–Candelon test

- Breitung and Candelon (2006): no causality at frequency ω is two linear restrictions on the coefficients of y in the x equation

$$M_{y\to x}(\omega) = 0 \iff \sum_{j=1}^{p} \psi_j \cos(j\omega) = 0 \quad \text{and} \quad \sum_{j=1}^{p} \psi_j \sin(j\omega) = 0$$

  - ψ_j: coefficient of y_{t-j} in the x equation; p: VAR order; an $F(2, T-2p-1)$ test at each ω, T: number of observations
  - with $p \leq 2$ the two restrictions fix all ψ_j, so the test is the same at every frequency
- BC show that the test extends to cointegrated VARs (long-run causality at $\omega = 0$); pointwise tests across frequencies are not a joint test

Does the euro area cause Romanian industry, and at which frequencies?



- VAR(3) in monthly industrial production growth (order by AIC, at least 3); left: Geweke measures; right: Breitung–Candelon F with its 5% critical value 3.03

ATS_ch11_causality

Interpreting frequency-domain causality

- Euro area $\rightarrow$ Romania: total Geweke measure 0.146 (integral of the curve: 0.145); Romania $\rightarrow$ euro area: 0.004
- The measure is 0.087 in the business-cycle band and 0.168 at periods under 6 months: past euro-area growth predicts Romanian growth mostly at short horizons
- The BC test rejects at 100% of frequencies for euro area $\rightarrow$ Romania and at no frequency in the other direction
- A small open economy follows its main market, not the reverse; the frequency profile says which kind of shock transmits fastest

Recap: Two series

- Coherence is an R^2 by frequency; it must be smoothed and tested against $1 - \alpha^{1/(L-1)}$
- Phase gives leads and lags, modulo 2π, only where coherence is significant; dynamic correlation keeps the sign
- Geweke decomposes Granger causality by frequency; Breitung–Candelon tests it with two linear restrictions

Band-pass filters

- Ideal filter for periods $[p_l, p_u]$: gain 1 on the frequencies $[\omega_1, \omega_2] = [2\pi/p_u, 2\pi/p_l]$, 0 elsewhere; its weights are infinitely many

$$b_0 = \frac{\omega_2 - \omega_1}{\pi}, \qquad b_j = \frac{\sin j\omega_2 - \sin j\omega_1}{\pi j}, \quad j = \pm 1, \pm 2, \ldots$$

- Baxter and King (1999): truncate at K lags and add a constant so that $\sum_{j=-K}^{K} a_j = 0$
  - a_j: the adjusted weights; the zero sum removes a unit root and a linear trend; symmetric, no phase shift
  - quarterly recommendation: BK(6, 32, $K = 12$); the price: K observations lost at each end; $a_0 = 0.28$
- Christiano and Fitzgerald (2003): minimise $\mathbb{E}[(y_t^{\text{ideal}} - \hat{y}_t)^2 \mid y_1, \ldots, y_n]$ assuming a random walk
  - y_t^{ideal}: the ideal band-pass component; asymmetric, time-varying weights that use the whole sample: no lost observations, but phase shifts near the ends
- The business-cycle band of 6–32 quarters (1.5–8 years) follows Burns and Mitchell, as adopted by Baxter and King (1999)

The Hodrick–Prescott filter



Edward C. Prescott, 2015

Photo: Narodowy Bank Polski (2015); CC BY-SA 2.0; Wikimedia Commons

- Hodrick and Prescott (1997): the trend balances fit against smoothness

$$\hat{\tau} = \arg\min_{\tau} \sum_t (y_t - \tau_t)^2 + \lambda \sum_t (\Delta^2 \tau_t)^2 = (I + \lambda D' D)^{-1} y$$

  - τ_t: trend; $\Delta^2 \tau_t$: its second difference; λ: smoothness penalty; D: the second-difference matrix; cycle $y_t - \hat{\tau}_t$

- Infinite-sample gain of the cycle: a high-pass filter

$$G(\omega) = \frac{4\lambda(1 - \cos\omega)^2}{1 + 4\lambda(1 - \cos\omega)^2}$$

  - the Kalman smoother of a local linear trend with signal-to-noise ratio $1/\lambda$ (Chapter 6); $\lambda = 1600$ for quarters; Ravn–Uhlig scaling $\lambda \propto s^4$ (s: observations per quarter): 129 600 for months, 6.25 for years Ravn and Uhlig (2002)

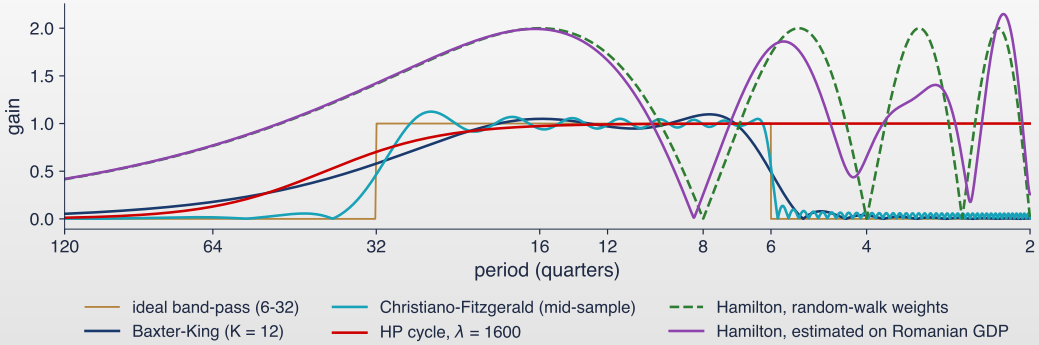
Hamilton's critique

- Hamilton (2018): "why you should never use the HP filter"
  - (1) applied to a random walk, HP produces a cycle with dynamics that are artefacts of the filter Cogley and Nason (1995)
  - (2) end-of-sample values are very different from the values the same filter gives once more data arrive (one-sided against two-sided)
  - (3) the λ implied by a statistical model fitted to the data is far from 1600
- Alternative: the **regression filter**, the cycle is the error of an h-step-ahead regression forecast

$$y_{t+h} = \beta_0 + \beta_1 y_t + \cdots + \beta_p y_{t-p+1} + v_{t+h}, \qquad \text{cycle}_{t+h} = \hat{v}_{t+h}$$

  - quarterly $h = 8$, $p = 4$; one-sided by construction; for a random walk $v_{t+h} = y_{t+h} - y_t$
- Replies: boosting the HP filter Phillips and Shi (2021); a modified Hamilton filter for real time Quast and Wolters (2022); the debate is about which gain you want

Gain functions of the filters



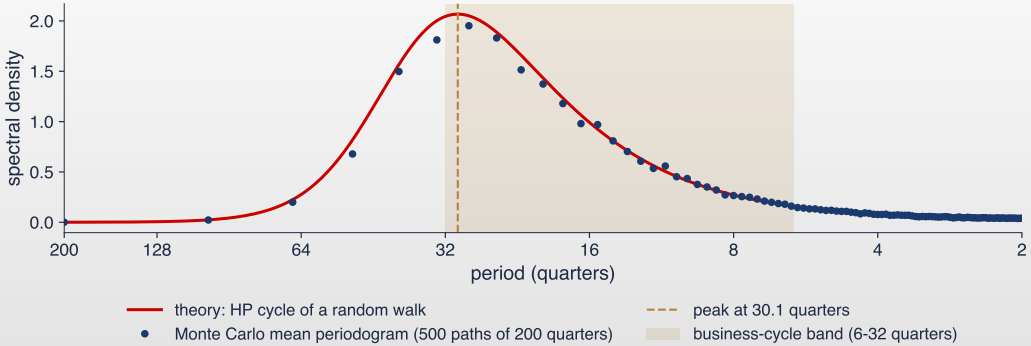
⊡ Gain from the level to the cycle, by period in quarters; Christiano–Fitzgerald: the weights in the middle of a 126-quarter sample; Hamilton: $h = 8$, $p = 4$

ATS_ch11_filters

Interpreting the gains

- HP is a high-pass filter: gain 0.70 at 32 quarters, 0.49 at 40 and still 0.16 at 60: the "HP cycle" contains cycles longer than 8 years
- BK ripples around 1 (maximum 1.10) and lets in 0.41 of the 40-quarter cycle; CF is closer to the ideal shape in mid-sample
- Hamilton's filter is not a band-pass: gain 2.00 at 16 quarters and 0.00 at 8 (for random-walk weights); the estimated Romanian weights (sum 0.97) give 1.99 and 0.26
- Every filter defines its own cycle: compare cycles only after comparing gains

The spurious cycle of the HP filter



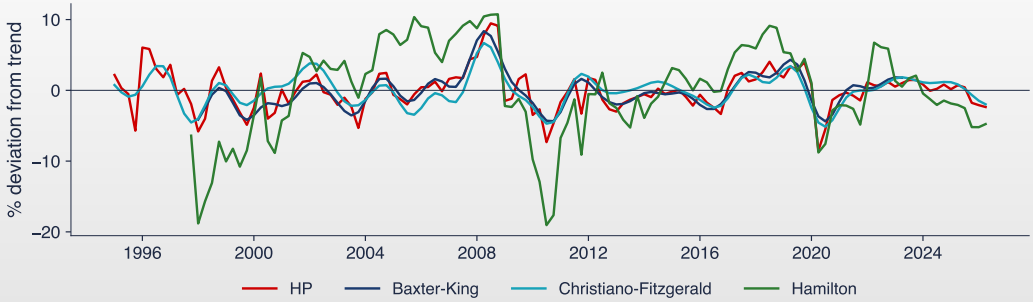
- Spectrum of the HP ($\lambda = 1600$) cycle of a random walk: $G(\omega)^2/(2\pi \cdot 2(1-\cos\omega))$, and the average periodogram of HP-filtered simulated random walks

ATS_ch11_filters

Interpreting the spurious cycle

- A random walk has no cycle, yet its HP cycle peaks at a period of 30.1 quarters (7.5 years); simulation: 28.6
- Derivation (Seminar 11, A1): maximise $u^3/(1+4\lambda u^2)^2$ in $u = 1-\cos\omega$: $u^* = \sqrt{3/(4\lambda)}$; replicates Cogley and Nason (1995)
- The peak sits inside the business-cycle band: "stylised facts" computed from HP cycles can be properties of the filter
- Ravn–Uhlig scaling keeps the peak near 7.5 years at monthly and annual frequency too (Seminar 11, A2)

Romanian business cycles by four filters



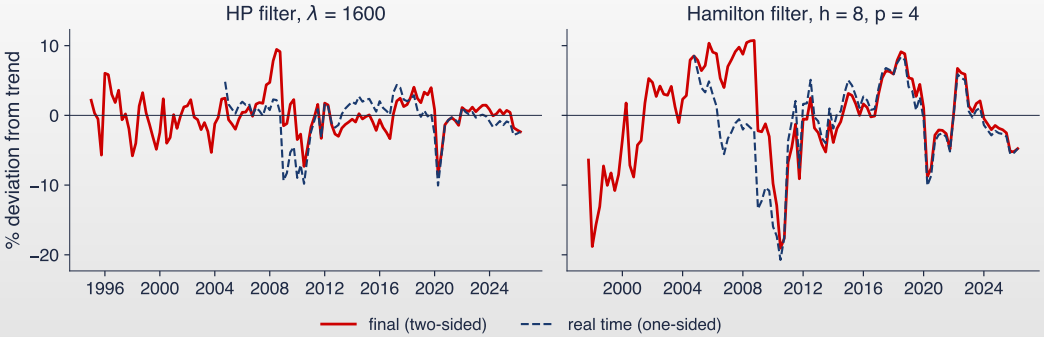
- 100 × log real GDP of Romania, 1995Q1 – 2026Q2; HP ($\lambda = 1600$), BK(6, 32, 12), CF(6, 32, random walk with drift), Hamilton ($h = 8$, $p = 4$)

ATS_ch11_filters

Interpreting the Romanian cycles

- All four place the 2008 peak (HP 9.5%, BK 8.4%) and the trough of 2010Q3 (HP -7.3%, BK -4.4%) at the same time
- Amplitudes differ: standard deviation HP 2.9, BK 2.5, CF 2.2, Hamilton 6.6; the Hamilton cycle is an 8-quarter forecast error, with a trough of -19.0%
- Correlations: BK–HP 0.84, BK–CF 0.84, HP–Hamilton 0.69, CF–Hamilton 0.49
- Today's position is uncertain: HP -2.4%, CF -2.0%, Hamilton -4.8% in 2026Q2; BK stops at 2023Q2

The end-point problem



- Romanian GDP: the cycle at each date computed with the data available at that date (real time) and with the full sample (final), from 2004Q4

ATS_ch11_filters

Interpreting the real-time revisions

- HP: root mean square revision 2.8 points, as large as the cycle itself (standard deviation 2.8); correlation of real-time and final cycles 0.57; sign wrong in 38% of quarters
- Hamilton: correlation 0.76, sign wrong in 24% of quarters; but the revision is 4.6 points, because the regression is re-estimated on short samples in a transition economy
- The real-time HP filter saw only a small part of the 2008 boom: the output gap that policy needed was the one it measured worst
- Report real-time properties of any gap estimate; the final estimate is not information that anyone had

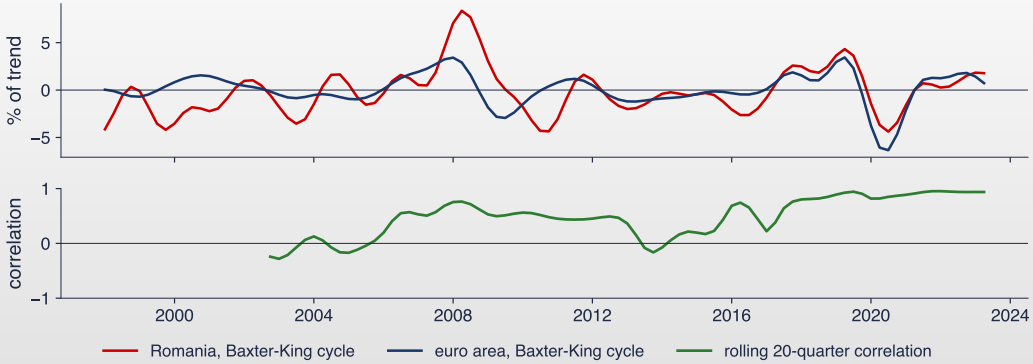
Measuring business-cycle synchronisation

- Correlation of filtered cycles; it depends on the filter (gain) and on a few large episodes
- **Concordance** Harding and Pagan (2002): the share of periods in which the two economies are in the same phase

$$C = \frac{1}{T} \sum_t \left[S_{xt} S_{yt} + (1 - S_{xt})(1 - S_{yt}) \right]$$

  - $S_{xt} = 1$ if x is in expansion (here: cycle above trend), 0 otherwise; under independence $\mathbb{E}C = p_x p_y + (1 - p_x)(1 - p_y)$, p_x, p_y: shares of expansion periods
- Dynamic correlation over the business-cycle band Croux, Forni and Reichlin (2001), which needs no filter
- Evidence for Central and Eastern Europe: a meta-analysis of 35 publications Fidrmuc and Korhonen (2006)
  - some countries already had high correlations with the euro area
  - the estimation method changes the correlation significantly

Romania and the euro area: synchronisation over time



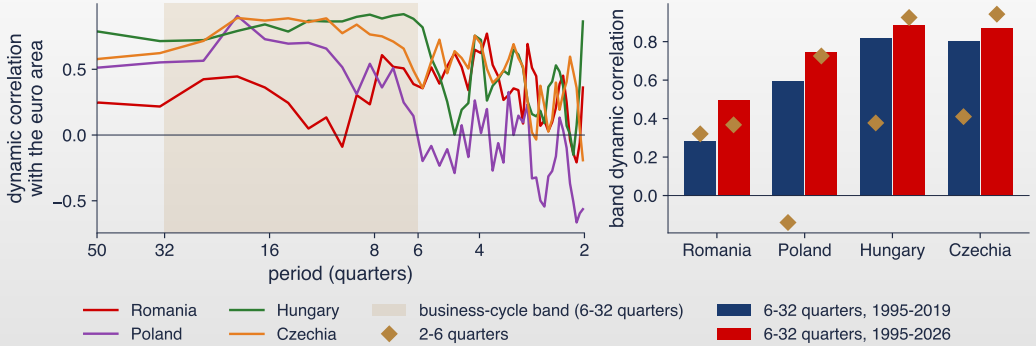
Baxter–King cycles of real GDP, 1998Q1 – 2023Q2, and their rolling 20-quarter correlation

ATS_ch11_filters

Interpreting synchronisation

- Correlation of the cycles: 0.54 over the whole sample; 0.34 before 2008, 0.47 in 2008–2012, 0.90 in 2013–2019, 0.96 from 2020
- Concordance 0.75 against 0.50 under independence: the two economies are above or below trend together most of the time
- The Romanian cycle is larger (standard deviation 2.5 against 1.7): synchronised in timing, not in amplitude
- The post-2020 correlation is driven by one common shock (the pandemic): a few quarters can make a decade look synchronised

Dynamic correlation with the euro area



⊡ Quarterly GDP growth, multitaper $NW = 3$; left: $\rho(\omega)$ for 1995–2019; right: band averages for 6–32 quarters (bars) and 2–6 quarters (diamonds), without and with 2020–2026

ATS_ch11_cross_spectrum

Interpreting the dynamic correlations

- Before 2020, business-cycle dynamic correlation: Hungary 0.81, Czechia 0.80, Poland 0.60, Romania 0.28
- Poland's short-run correlation is -0.14: its growth co-moved with the euro area only through the business cycle
- With 2020–2026 every number rises (Romania 0.50, Poland 0.75): the pandemic quarter dominates every cross-periodogram
- Report synchronisation with and without extreme common shocks; robust (trimmed or rank-based) spectra are a research topic

Recap: Filters and the business cycle

- A cycle is defined by a gain: HP is high-pass, BK and CF approximate a band-pass, Hamilton's regression filter is neither
- HP creates a 30-quarter cycle from a random walk and is unreliable in real time
- Romania is synchronised with the euro area in timing since 2010, with a larger amplitude

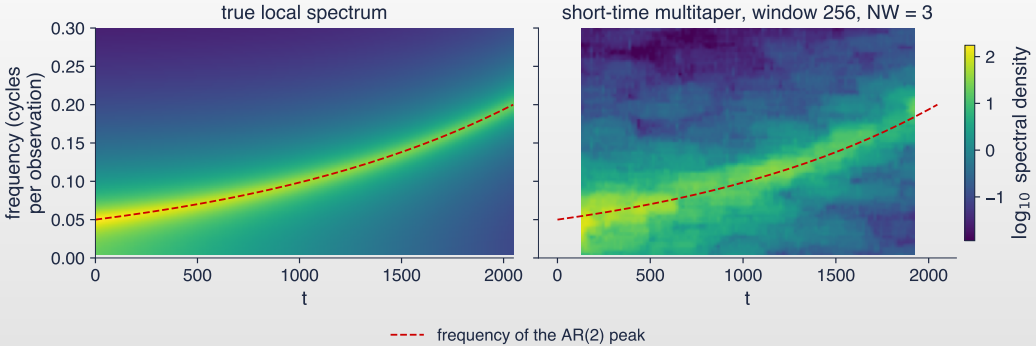
Evolutionary spectra and local stationarity

- Priestley (1965): Cramér's representation with a slowly changing amplitude

$$X_t = \int A_t(\omega) e^{i\omega t}\, dZ(\omega), \qquad f_t(\omega) = |A_t(\omega)|^2 f(\omega)$$

  - $A_t(\omega)$: time-varying amplitude; $f_t(\omega)$: the **evolutionary spectrum** at time t
- Dahlhaus (1997): **locally stationary** processes $X_{t,T}$ with $A(t/T, \omega)$ smooth in rescaled time $u = t/T$
  - T: sample size; a time-varying spectrum $f(u, \omega)$ that can be estimated consistently by a tapered spectrum on a moving window (spectrogram)
- **Uncertainty principle**: a window of length L resolves frequencies only to about $1/L$; long windows blur time, short windows blur frequency
- Wavelets (next sections) let the window length change with the frequency: short for high frequencies, long for low ones

A time-varying spectrum: spectrogram of a simulated series



- $X_t = 2r\cos(\theta_t)X_{t-1} - r^2 X_{t-2} + \varepsilon_t$, $r = 0.95$, peak period moving from 20 to 5 over 2048 observations; short-time multitaper on windows of 256 (step 16)

ATS_ch11_evolutionary

Interpreting the spectrogram

- The spectrogram follows the moving peak: a single spectrum of the whole sample would show a smeared band from 0.05 to 0.2 cycles
- Resolution $2NW/L = 0.023$ cycles: fine at high frequencies, coarse for the long periods at the start
- The first and last 128 observations have no estimate: every time–frequency method loses the edges (the cone of influence of wavelets)
- Economic examples: changing business-cycle length, the shortening of trading cycles, regime changes in volatility

Singular spectrum analysis

- Embed $x_1, \ldots, x_n$ in the $L \times (n - L + 1)$ trajectory (Hankel) matrix X; SVD $\mathrm{X} = \sum_i \sigma_i \mathrm{u}_i \mathrm{v}_i'$ Vautard and Ghil (1989)
- Group eigentriples (trend: slowly varying u_i; oscillations: pairs with equal σ_i and shifted sines) and reconstruct each group by averaging anti-diagonals
- Data-adaptive basis instead of fixed sinusoids or wavelets; no model, no stationarity assumption; inference is less developed
- A code sketch (ssa) is in the Quantlet engine of this chapter; it is a natural benchmark for wavelet multiresolution

Recap: Spectra that change over time

- Priestley and Dahlhaus give a meaning to a time-varying spectrum
- A fixed window trades time against frequency resolution
- Wavelets adapt the window to the frequency; SSA adapts the basis to the data

From sinusoids to wavelets



Ingrid Daubechies, 2005

Photo: own work of the uploader (2005); public domain; Wikimedia Commons

- A **wavelet** ψ: a short wave, localised in time and frequency Grossmann and Morlet (1984)
  - $\int \psi = 0$, $\int \psi^2 = 1$; admissible if $C_\psi = \int |\hat{\psi}(\omega)|^2/|\omega|\, d\omega < \infty$, with $\hat{\psi}$ the Fourier transform of ψ
- Continuous wavelet transform (CWT): correlate the series with stretched and shifted copies of ψ

$$W(s,\tau) = \int x(t)\, \frac{1}{\sqrt{s}}\, \psi^*\Big(\frac{t-\tau}{s}\Big)\, dt$$

  - $s > 0$: scale (inversely related to frequency); τ: location in time; ψ^*: complex conjugate
- Compactly supported orthonormal wavelets Daubechies (1988) and the pyramid algorithm Mallat (1989) give the DWT: n coefficients for n observations; applications: Crowley (2007); Gençay, Selçuk and Whitcher (2002); Aguiar-Conraria and Soares (2014)

DWT, MODWT and multiresolution (1/2)

- Two filters of length L Percival and Walden (2000)

$$\text{scaling (low-pass): } g_l, \qquad \text{wavelet (high-pass): } h_l = (-1)^l g_{L-1-l}$$

  - Haar $L = 2$; Daubechies least-asymmetric LA(8) $L = 8$; level j captures periods $[2^j, 2^{j+1}]$: an **octave** band; J levels plus a smooth

- **MODWT** (maximal overlap DWT): no downsampling, rescaled filters

$$\tilde{h} = h/\sqrt{2}, \quad \tilde{g} = g/\sqrt{2}, \qquad \tilde{W}_{j,t} = \sum_l \tilde{h}_{j,l} X_{t-l \bmod n}$$

  - $\tilde{h}_{j,l}$: the level-j wavelet filter; $\tilde{W}_{j,t}$: wavelet coefficient of level j at time t; mod$n$: the sample is treated as circular

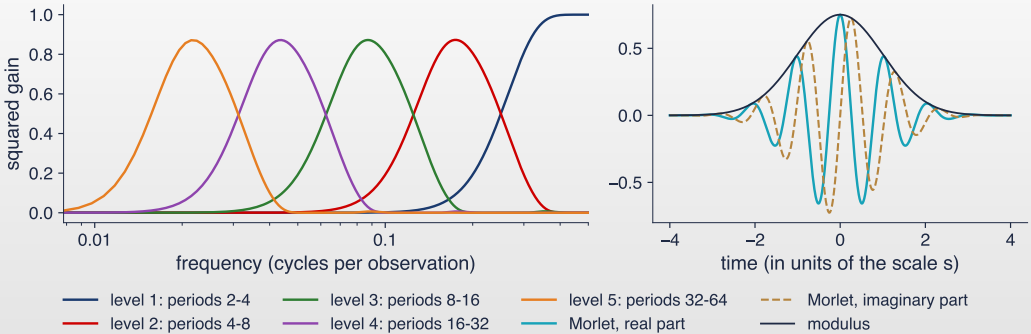
DWT, MODWT and multiresolution (2/2)

- MODWT properties
  - any sample size, shift-invariant (moving the start date does not change the coefficients), $J \times n$ coefficients
  - energy preserved: $\|X\|^2 = \sum_j \|\tilde{W}_j\|^2 + \|\tilde{V}_J\|^2$, with $\tilde{V}_J$ the scaling (smooth) coefficients
- **Multiresolution analysis (MRA)**: the series is the sum of its details and a smooth

$$X_t = \sum_{j=1}^{J} \mathcal{D}_{j,t} + \mathcal{S}_{J,t}$$

  - $\mathcal{D}_j$: the inverse MODWT of level j alone; $\mathcal{S}_J$: the smooth; zero-phase, aligned with the data
- Boundary: the first $L_j - 1 = (2^j - 1)(L - 1)$ coefficients wrap around the sample (circular filtering) and are excluded from estimation

Wavelet filters and the Morlet wavelet



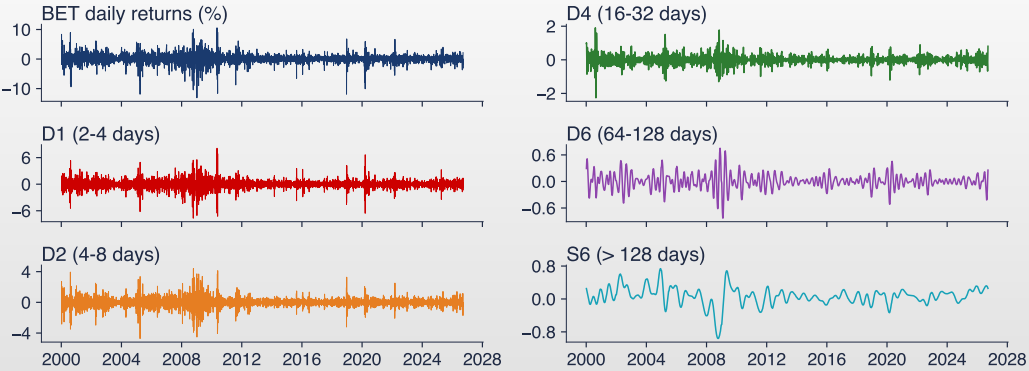
- Left: squared gains of the MODWT LA(8) filters of levels 1–5; right: the Morlet wavelet $\psi_{\mathbf{0}}(\eta) = \pi^{-\mathbf{1/4}} e^{i\omega_{\mathbf{0}}\eta} e^{-\eta^{\mathbf{2}}/\mathbf{2}}$, $\omega_{\mathbf{0}} = 6$

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Interpreting the wavelet filters

- Each MODWT level is an approximate band-pass filter for one octave; the bands overlap, so leakage between neighbouring levels is part of the design
- The bandwidth grows with frequency: fine frequency resolution for long cycles, fine time resolution for short ones
- Morlet is complex: its modulus gives amplitude, its argument gives phase; Fourier period $= 1.033\times$ scale for $\omega_0 = 6$
- Orthonormal filters (MODWT) serve variance decomposition and correlation by scale; the Morlet CWT serves time–frequency maps and phase

The BET return decomposed by scale



- MODWT multiresolution analysis, LA(8), $J = 6$, daily BET returns 2000–2026 (6 670 days); four of the six details and the smooth

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Interpreting the multiresolution analysis

- Energy shares of levels 1–6: 43.0, 26.6, 15.0, 6.8, 3.5, 2.3%; white noise would give 50.0, 25.0, 12.5, 6.2, 3.1, 1.6%
- Less energy at the shortest scale and more at the longer ones than white noise: positive autocorrelation of BET returns at short horizons
- Volatility clustering appears at every scale in 2008 and 2020, but the slow components ($\mathcal{D}_6$, $\mathcal{S}_6$) move only in the 2008–2009 crisis
- The details add up exactly to the series: a lossless, zero-phase decomposition

Wavelet variance, correlation and beta by scale (1/2)

- **Wavelet variance**: the variance of the level-j coefficients; the levels add up to the variance of the series Percival (1995)

$$\nu_X^2(\tau_j) = \mathrm{Var}\tilde{W}_{j,t}, \qquad \sum_j \nu_X^2(\tau_j) = \mathrm{Var}X$$

  - $\tau_j = 2^{j-1}$: the scale of level j; an octave-band version of the spectrum; white noise: $\nu^2(\tau_j) = \sigma^2/2^j$, halving at each level
  - unbiased estimator: average of $\tilde{W}_{j,t}^2$ over the $M_j = n - L_j + 1$ non-boundary coefficients; $\eta\hat{\nu}^2/\nu^2 \approx \chi_\eta^2$, $\eta = \max(M_j/2^j, 1)$ equivalent degrees of freedom

Wavelet variance, correlation and beta by scale (2/2)

- **Wavelet correlation** Whitcher, Guttorp and Percival (2000) and **wavelet beta** Gençay, Selçuk and Whitcher (2005)

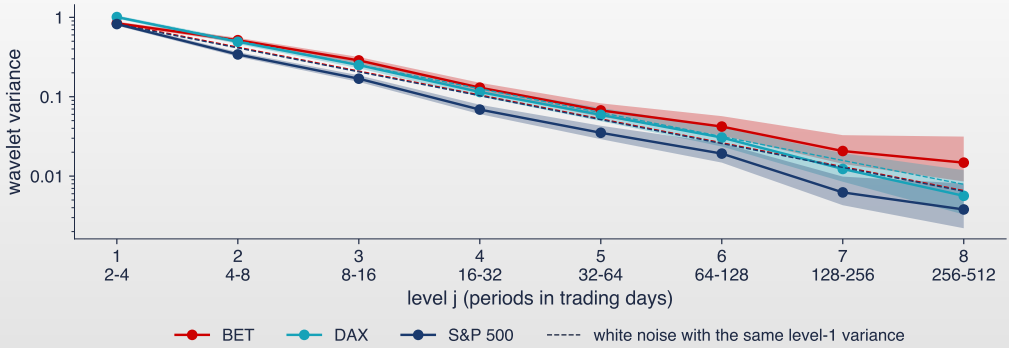
$$\rho_{XY}(\tau_j) = \frac{\mathrm{Cov}(\tilde{W}_j^X, \tilde{W}_j^Y)}{\nu_X(\tau_j)\nu_Y(\tau_j)}$$

$$\beta(\tau_j) = \frac{\mathrm{Cov}(\tilde{W}_j^r, \tilde{W}_j^m)}{\mathrm{Var}\tilde{W}_j^m}$$

  - $\tilde{W}_j^r$, $\tilde{W}_j^m$: coefficients of an asset return and of the market return; $\beta(\tau_j)$: systematic risk at the investment horizon τ_j
  - 95% interval for the correlation: $\tanh\left(\tanh^{-1}\hat{\rho} \pm z/\sqrt{M_j/2^j - 3}\right)$, $z = 1.96$

- Effective sample size falls by half at each level: long-scale estimates are wide even with 25 years of daily data

Wavelet variance of three equity indices



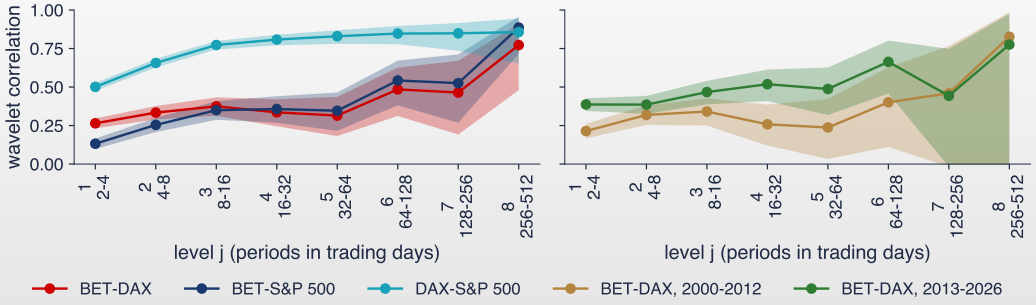
- Daily log returns 2000–2026, MODWT LA(8), levels 1–8 with 95% intervals; dashed: white noise with the same level-1 variance

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Interpreting the wavelet variances

- Ratio of the variance at level 6 (64–128 days) to the white-noise line: BET 1.60, DAX 0.97, S&P 500 0.75
- BET has more long-horizon variance than its daily variance implies (level 8: 2.25): momentum, a thin market; the S&P 500 has less (0.59): mean reversion of daily moves
- This is the variance-ratio test of market efficiency, done scale by scale with valid intervals
- Risk measured on daily data and scaled by $\sqrt{h}$ understates BET risk at long horizons

Co-movement by scale: BET, DAX and S&P 500



- Daily returns on 6 433 common trading days, MODWT LA(8); left: three pairs, 2000–2026; right: BET–DAX in two halves; 95% Fisher-z intervals

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Interpreting the wavelet correlations

- BET–DAX rises from 0.26 (2–4 days) to 0.77 (256–512 days); the daily correlation (0.31) understates long-run integration
- BET–S&P 500 at level 1 is only 0.13, DAX–S&P 500 0.50: New York closes after Europe, so daily co-movement shows up a day late; it vanishes at longer scales
- BET–DAX at 8–16 days: 0.34 in 2000–2012 against 0.47 in 2013–2026 (level 3): integration has increased at short and medium scales
- Wavelet beta of BET on DAX: 0.24 at level 1, 0.56 at level 6: the diversification benefit of the BET shrinks with the horizon (Gençay, Selçuk and Whitcher (2005); Rua and Nunes (2009))

Recap: Wavelets: decomposition by scale

- The MODWT splits variance and covariance into octave bands, for any n and without phase shift
- Wavelet variance is a scale-by-scale variance-ratio test; wavelet correlation and beta show integration by horizon
- Drop boundary coefficients and use equivalent degrees of freedom for intervals

The Morlet transform and its significance (1/2)

- Torrence and Compo (1998): the CWT computed by FFT on a grid of scales

$$W_n(s) = \sum_k \hat{x}_k \, \hat{\psi}^*(s\omega_k) \, e^{i\omega_k n \delta t}, \qquad s_j = s_0 2^{j\delta j}$$

  - $\hat{x}_k$: the DFT of the series at frequency ω_k; $\hat{\psi}$: the Fourier transform of the Morlet wavelet; δt: time step; s_0: smallest scale; δj: scale step in octaves
  - wavelet power $|W_n(s)|^2$: the variance of the series at time n and scale s
- **Cone of influence** (COI): where the wavelet at scale s reaches the edge, within $\sqrt{2}\,s$ of either end
  - results inside the COI are biased and are not interpreted

The Morlet transform and its significance (2/2)

- Null of red noise, an AR(1) with lag-1 autocorrelation a: the power is a scaled χ^2_2

$$\frac{|W_n(s)|^2}{\sigma^2} \sim P_k \frac{\chi^2_2}{2}, \qquad P_k = \frac{1-a^2}{1+a^2-2a\cos(2\pi k/N)}$$

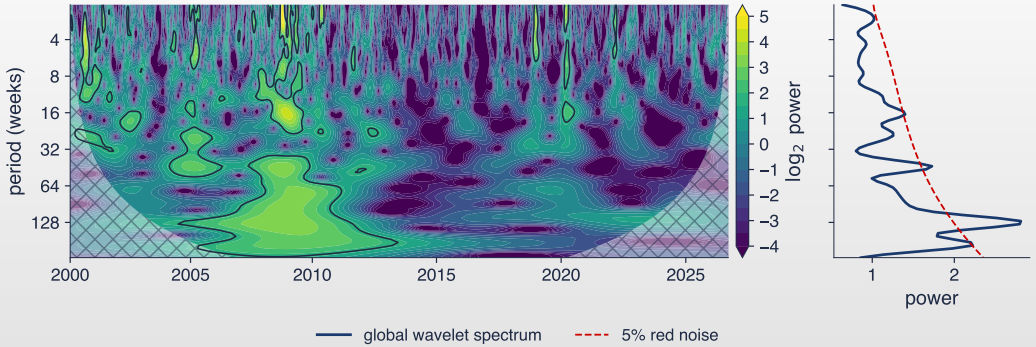
  - σ^2: variance of the series; P_k: the red-noise spectrum at the Fourier index k matching scale s; N: number of observations

- Global wavelet spectrum: the time average of the power, a smoothed Fourier spectrum

$$\bar{W}^2(s) = \frac{1}{n}\sum_n |W_n(s)|^2, \qquad \nu = 2\sqrt{1+\left(\frac{n\delta t}{2.32\,s}\right)^2} \text{ degrees of freedom}$$

- Rectify the power by $1/s$ when comparing scales Liu, Liang and Weisberg (2007); the pointwise test runs at thousands of points: expect 5% false patches Maraun and Kurths (2004)

Wavelet power of the BET



- Weekly BET returns (standardised), 1 379 weeks 2000–2026, Morlet $\omega_0 = 6$, $\delta j = 1/12$; black contours: 5% red-noise test; hatched: cone of influence; right: global wavelet spectrum

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Interpreting the wavelet power

- Maximum power at a period of 2.6 weeks on 17 October 2008: the Lehman weeks; significant power in 2007–2011 at periods of 32 to 128 weeks
- Lag-1 autocorrelation 0.04: the red-noise null is almost white; 11% of the reliable area is significant at 5%
- Power in returns is variance, not a cycle: the map shows when volatility was concentrated at which horizons
- The global spectrum stays inside the red-noise band: no persistent cycle in BET returns, as efficiency predicts

Wavelet coherence and phase (1/2)

- Cross-wavelet transform and **squared wavelet coherence** Torrence and Webster (1999), Grinsted, Moore and Jevrejeva (2004)

$$W_n^{xy}(s) = W_n^x(s) W_n^{y*}(s)$$

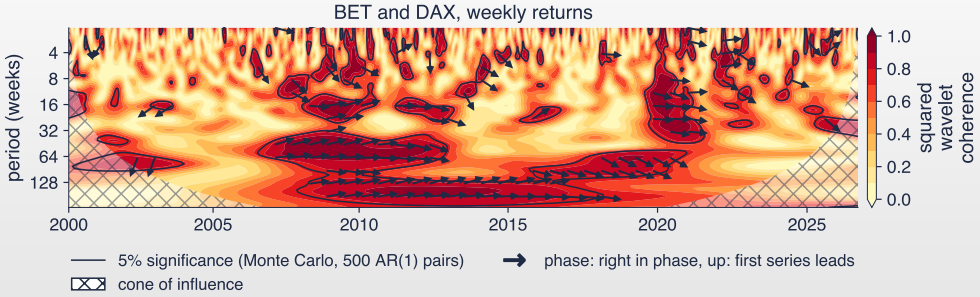
$$R_n^2(s) = \frac{|\mathcal{S}(s^{-1} W_n^{xy}(s))|^2}{\mathcal{S}(s^{-1}|W_n^x(s)|^2)\,\mathcal{S}(s^{-1}|W_n^y(s)|^2)} \in [0,1]$$

  - $\mathcal{S}$: smoothing, Gaussian in time (width s) and a boxcar of 0.6 octaves in scale; $R_n^2(s)$: a local R^2 at time n and scale s
  - without smoothing $R^2 \equiv 1$, as for the raw coherence: smoothing is what makes it an estimator

Wavelet coherence and phase (2/2)

- Phase $\arg \mathcal{S}(s^{-1} W^{xy})$, shown as arrows
  - right = in phase, left = anti-phase, up = x leads; lead in time = phase $\times$ period$/2\pi$
- Significance by **Monte Carlo** Grinsted, Moore and Jevrejeva (2004)
  - simulate pairs of independent AR(1) series with the data's lag-1 autocorrelations; the 95th percentile of R^2 at each scale is the 5% threshold
- Extensions: partial and multiple wavelet coherence, phase differences with confidence bands Aguiar-Conraria and Soares (2014); monetary policy Aguiar-Conraria, Azevedo and Soares (2008); energy Vacha and Barunik (2012)

BET and DAX in time and frequency



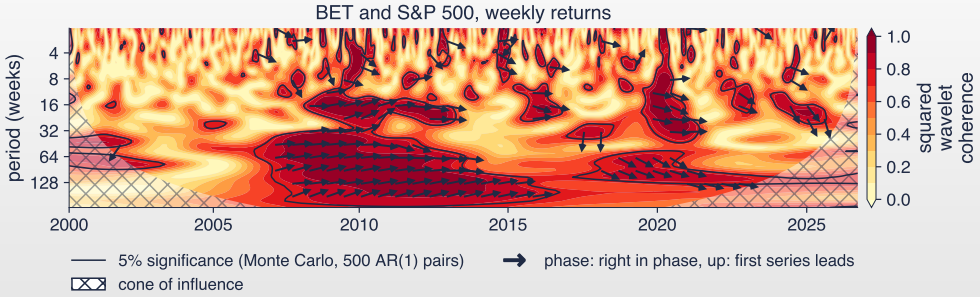
- Weekly returns 2000–2026; contours: 5% significance from 500 AR(1) pairs; arrows only where significant

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Interpreting BET–DAX coherence

- 23% of the reliable area is significant (null 95th percentile 7.7%): the co-movement is real, but concentrated
- Average R^2 at 8–64 weeks: 0.37 in 2000–2006, 0.76 in 2008–2009, 0.36 in 2014–2019, 0.77 in 2020–June 2021
- Phase near zero in crises (0.03 rad in 2008–2009), negative before 2007 (-0.97 rad): the DAX led the BET by weeks when the Bucharest market was young
- Crises raise coherence: is it contagion or only higher volatility? Correlations rise mechanically with volatility Forbes and Rigobon (2002), and so does R^2

BET and S&P 500 in time and frequency



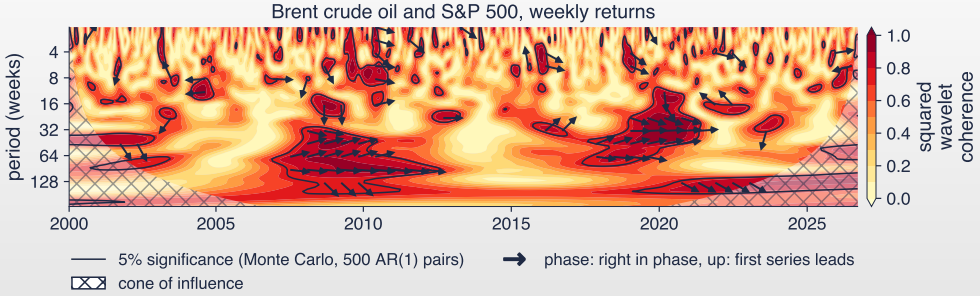
- Weekly returns 2000–2026; Monte Carlo significance as before

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Interpreting BET–S&P 500 coherence

- Significant area 26% (null 95th percentile 7.7%); R^2 at 8–64 weeks 0.76 in 2008–2009 and 0.48 in 2014–2019
- In 2000–2006 the phase is -1.24 rad: the S&P 500 led by about a fifth of the cycle; in crises the markets move in phase
- Weekly data remove the one-day lag of the daily wavelet correlation: frequency choice matters for lead–lag claims
- Our picture matches the international evidence: co-movement is strongest at low frequencies and in crises Rua and Nunes (2009)

Oil and stocks



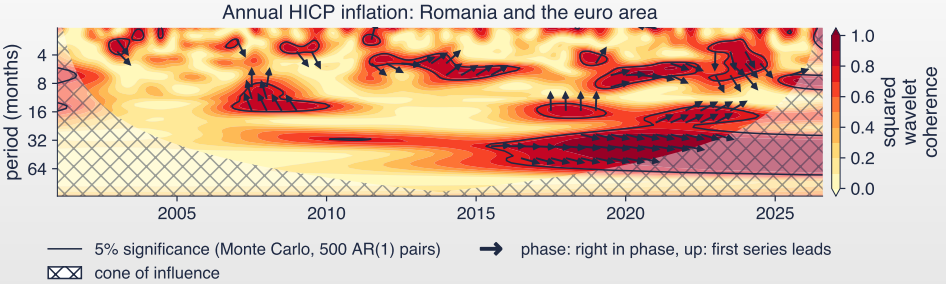
- Weekly returns of Brent crude oil (FRED) and the S&P 500, 2000–2026; Monte Carlo significance from 500 AR(1) pairs

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Interpreting oil–stock coherence

- Significant area 15% (null 95th percentile 7.1%): oil and stocks are linked only in episodes
- Before 2007: R^2 0.41 with phase -1.86 rad, neither in phase nor in anti-phase: a weak lead–lag link; 2008–2010: 0.70, in phase (-0.09 rad): a common demand shock
- 2020–2021: 0.58 in phase; 2022–2023: phase -2.31 rad, closer to anti-phase: the energy-price shock of the war in Ukraine
- The sign of the oil–stock link depends on the type of shock Kilian and Park (2009); wavelet phase shows the switch without identifying it

Inflation in Romania and the euro area



- Annual HICP inflation (monthly), January 2001 – August 2026; Monte Carlo significance against AR(1) pairs with lag-1 autocorrelation 0.97

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Interpreting inflation co-movement

- Romania–euro area: R^2 at 24–64 months 0.23 in 2002–2012, 0.85 from 2015; significant area 17% (null 95th percentile 8.7%)
- Poland: 0.29 and 0.91; simple correlations 0.34 (Romania) and 0.86 (Poland): Romanian disinflation to 2012 was domestic, the cycle since 2015 is common
- 2021–2024 surge, 12–48 months: R^2 0.78, phase 0.14 rad: no detectable lead, a common energy shock
- Annual rates overlap twelve months: short periods are smoothed away by construction; only cycles longer than a year are informative

Recap: Wavelet coherence

- Wavelet coherence is a smoothed, time-local coherence; without smoothing it is identically one
- Test it by Monte Carlo against AR(1) pairs; read phase only inside significant regions and outside the cone of influence
- Equity co-movement peaks in crises and at long horizons; oil–stock phase flips with the type of shock

An open question

- Is the rise of stock-market coherence in crises contagion, or the mechanical effect of higher volatility on a time-local R^2?
  - formal: H_0: the wavelet coherence of BET and DAX in the pre-registered crisis windows equals its value under a model with the same time-varying volatilities and constant dependence
  - falsified if the crisis coherence exceeds the 95th percentile of a bootstrap from a DCC-GARCH with constant correlation (Chapter 8), at pre-registered scales
- Why it matters: diversification fails when it is needed; the Forbes–Rigobon correction exists for correlations, not for wavelet coherence
  - literature to start from: Forbes and Rigobon (2002), Rua and Nunes (2009), Maraun and Kurths (2004), Schulte (2016), Aguiar-Conraria and Soares (2014)

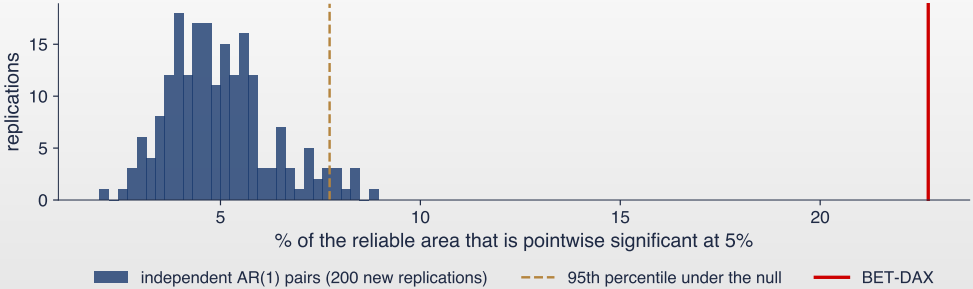
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
  - **literature**: `List peer-reviewed papers since 2009 that test wavelet coherence between stock markets against a volatility-matched null; give DOIs.` Then check every DOI on Crossref
  - **hypothesis**: `Derive the bias of the smoothed wavelet coherence when both series have a common GARCH volatility and constant correlation.`
  - **code and replication**: ask for the Grinsted smoothing operator, then reproduce a known case first (two AR(1) series: 5% significant area)
  - **critique**: `Act as a hostile referee: list the ways a significant wavelet-coherence island can arise without any change in dependence.`
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

What the human checks

- Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- Conventions: angular frequency or cycles, 2π factors, period $= 1.033\times$ scale for Morlet, the sign convention of the phase
- Coherence is smoothed; its threshold and the Monte Carlo null are stated; nothing is read inside the cone of influence
- Pointwise significance is not areawise significance: count how many islands the null produces
- Filters: gain stated; real-time and final cycles separated; results with and without extreme common shocks

Mini-case: spurious significance in wavelet coherence maps



- Share of the reliable time–period area of a wavelet-coherence map that is pointwise significant at 5%: 200 new pairs of independent AR(1) series with the BET and DAX autocorrelations, and the observed BET–DAX map
- Null: mean 5.0%, 95th percentile 7.7%, maximum 9.0%; 100% of null maps contain significant islands; BET–DAX: 22.7%. An AI summary that reads every island as a contagion episode is wrong; the overall excess is real

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Project idea

- **Contagion or volatility? Wavelet coherence of Central and Eastern European stock markets under a volatility-matched null**
  - replicate: the Monte Carlo test of Grinsted, Moore and Jevrejeva (2004) and the international co-movement maps of Rua and Nunes (2009) on BET, WIG20, BUX, PX against the DAX
  - extend: a DCC-GARCH bootstrap null (constant dependence, time-varying volatility); areawise tests Schulte (2016); partial coherence controlling for the S&P 500 Aguiar-Conraria and Soares (2014)
  - pre-register: markets, sampling frequency, crisis windows, scales, the null model and the number of replications
- Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- The spectrum decomposes variance by frequency; filters act on it through their squared gain
- Estimation is a bias–variance choice; multitaper gives low leakage, stable variance and honest bands
- Coherence, phase, dynamic correlation and Breitung–Candelon tests describe co-movement and causality by frequency
- A business cycle is whatever the filter's gain lets through; HP adds spurious cycles and fails in real time
- Wavelets localise variance and co-movement in time and scale; significance needs Monte Carlo and areawise thinking

Self-assessment

Questions

- Why is the periodogram inconsistent although it is asymptotically unbiased?
- How does the optimal bandwidth grow with n for a kernel with $q = 2$?
- Why is raw (unsmoothed) coherence always equal to one?
- Where does the peak of the HP cycle of a random walk come from?
- Why is the MODWT preferred to the DWT for wavelet variance?

Next:

- Machine learning and deep learning for time series
- global models, leakage-free validation, deep architectures

Appendix: the distribution of the periodogram

- Gaussian white noise: $a_j = \sqrt{2/n}\sum_t x_t \cos(\omega_j t)$, $b_j = \sqrt{2/n}\sum_t x_t \sin(\omega_j t)$ are i.i.d. $N(0, \sigma^2)$ for $0 < \omega_j < \pi$ (orthogonality of the Fourier basis)
- $I(\omega_j) = (a_j^2 + b_j^2)/(4\pi) = \frac{\sigma^2}{2\pi} \cdot \frac{\chi_2^2}{2}$: mean $f = \sigma^2/(2\pi)$, variance f^2, whatever n
- Linear process: $d_X(\omega_j) = \Psi(e^{-i\omega_j}) d_\varepsilon(\omega_j) + o_p(1)$, so $I_X(\omega_j) \approx |\Psi(e^{-i\omega_j})|^2 I_\varepsilon(\omega_j) = f(\omega_j)\chi_2^2/2$
- Averaging L such ordinates: mean f, variance f^2/L; this is the logic of every smoothed and multitaper estimator

Back

Appendix: two filter facts

- **Coherence is filter-invariant**: $\tilde{x} = A(L)x$, $\tilde{y} = B(L)y$ give $f_{\tilde{x}\tilde{y}} = A(\omega)\overline{B(\omega)}f_{xy}$, $f_{\tilde{x}} = |A|^2 f_x$, $f_{\tilde{y}} = |B|^2 f_y$, so $\kappa^2_{\tilde{x}\tilde{y}} = \kappa^2_{xy}$ where $A, B \neq 0$
- The phase changes by $\arg A - \arg B$: prewhitening with different filters shifts the phase; use the same filter on both series
- **HP cycle of a random walk**: $\Delta y = \varepsilon$, so $f_c(\omega) = G(\omega)^2\sigma^2/(2\pi \cdot 2(1 - \cos\omega))$; with $u = 1 - \cos\omega$ this is proportional to $u^3/(1 + 4\lambda u^2)^2$
- Setting the derivative to zero: $3(1 + 4\lambda u^2) = 16\lambda u^2$, so $u^* = \sqrt{3/(4\lambda)}$ and the period $2\pi/\arccos(1 - u^*)$

Back

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